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Cite this: *Org. Chem. Front.*, 2020, **7**, 3399

Phosphine-catalyzed [3 + 2] cycloadditions of trifluoromethyl enynes/enediynes with allenoates: access to cyclopentenes containing a CF_3-substituted quaternary carbon center†

Shu-Jie Chen,[a] Guo-Shu Chen,[a] Jia-Wei Zhang,[b] Zhao-Dong Li,[c] Yu-Lei Zhao[d] and Yun-Lin Liu*[a]

Received 7th July 2020,
Accepted 11th September 2020

DOI: 10.1039/d0qo00807a

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Herein, the first use of trifluoromethyl-substituted enynes/enediynes as non-classical electron-deficient olefins for phosphine-catalyzed [3 + 2] cycloaddition with allenoates is reported. This reaction occurs with excellent regioselectivity under mild conditions, affording alkyne- and diyne-tethered cyclopentene derivatives containing a CF_3-substituted quaternary carbon center in moderate to good yields.

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Five-membered carbocycles, including cyclopentenes and cyclopentanes, are the key structural moieties of a number of pharmaceuticals and bioactive natural and non-natural products (Fig. 1).[1] As a result, much attention has been devoted to developing efficient strategies for the construction and functionalization of cyclopentenes and cyclopentanes with high structural diversity.[2] On the other hand, the trifluoromethyl (CF_3) group plays important roles in the pharmaceutical and agrochemical industries because trifluoromethylated compounds often display improved physicochemical and biological properties compared to their non-fluorinated counterparts.[3] For example, incorporating the trifluoromethyl group into five-membered carbocycles has been found to impart beneficial effects (*e.g.* bioavailability and metabolic stability) on many bioactive molecules (Fig. 1, bottom).[1e,f] In this context, the development of methods for the expedient synthesis of CF_3-bearing five-membered carbocycles is highly desirable.

Among the existing methodologies for the synthesis of functionalized cyclopentenes, phosphine-catalyzed [3 + 2] cycloaddition of allenoates with electron-deficient olefins first reported by Lu's group in 1995 has shown notable efficiency.[4] Various olefins activated by one or more electron-withdrawing groups, such as carbonyl,[5] phosphonyl,[6] sulfonyl,[7] cyano,[8] and nitro groups,[9] have been successfully applied to these cycloaddition processes (Scheme 1a). In contrast, trifluoromethyl alkenes were less explored for this transformation. To the best of our knowledge, there have been only two reports on the use of trifluoromethyl alkenes as partners in Lu's [3 + 2] cycloaddition with allenes. Zhang's group reported chiral phosphine-catalyzed asymmetric [3 + 2] cycloadditions of γ-substituted allenoates with β-trifluoromethyl enones, which provided a facile access to highly functional trifluoromethylated cyclopentenes.[10] Loh and coworkers have demonstrated two examples of phosphine-catalyzed [3 + 2] cycloaddition of β-trifluoromethyl acrylates with aryl allenones (Scheme 1b).[11] In both reports, the CF_3 group only served as a second electron-withdrawing group and the application of trifluoromethyl olefins without strong electron-withdrawing π-acceptors in

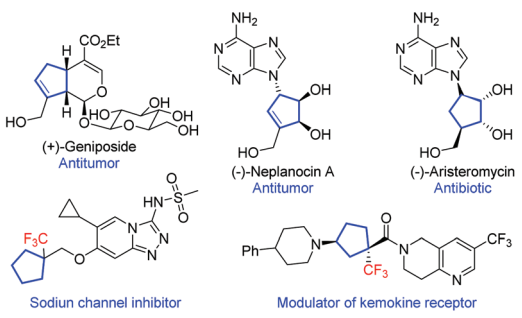
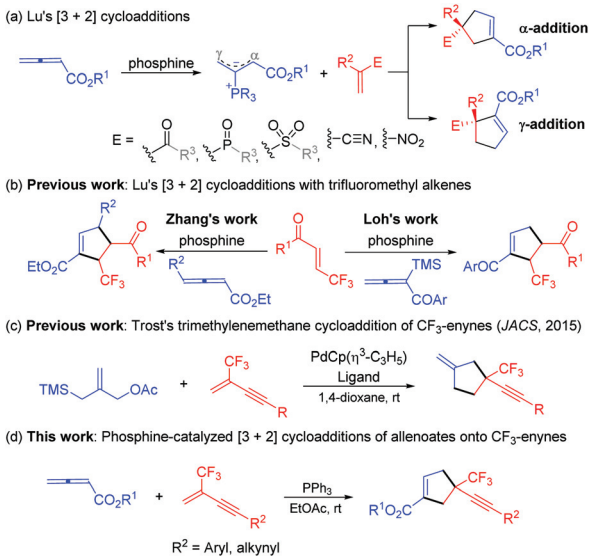


Fig. 1 Selected biologically active compounds containing cyclopentene or cyclopentane scaffolds.

[a]*School of Chemistry and Chemical Engineering, Guangzhou University, Guangzhou, 510006, China. E-mail: ylliu@gzhu.edu.cn*
[b]*College of Materials Science and Engineering, Hunan University, Changsha, 410082, China*
[c]*Department of Applied Chemistry, College of Materials and Energy, South China Agricultural University, Guangzhou, 510642, China*
[d]*School of Chemistry and Chemical Engineering, Qufu Normal University, Qufu, 273165, China*

†Electronic supplementary information (ESI) available. CCDC 2004999. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0qo00807a

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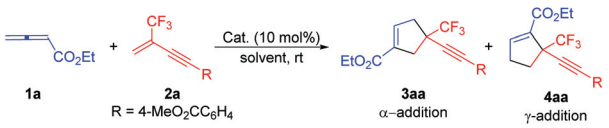
Scheme 1 Phosphine-catalyzed [3 + 2] annulation reaction of allenoates with alkenes.

Lu's [3 + 2] cycloaddition has not been realized so far. In fact, only a palladium-catalyzed [3 + 2] cycloaddition of CF_3-substituted 1,3-enynes with trimethylenemethane (TMM) was reported by Trost and coworkers in 2015, furnishing a facile access to cyclopentenes bearing a CF_3-substituted quaternary carbon center (Scheme 1c).[12]

Recently, we found that CF_3-substituted enynes/enediynes could undergo Michael-initiated ring-closure reactions with sulfur ylides, furnishing alkyne- and diyne-tethered cyclopropanes bearing trifluoromethyl-substituted all-carbon quaternary centers.[13] With our ongoing research interest in annulation reactions[14] and the development of practical syntheses of fluorine-containing compounds,[15] we wondered whether the CF_3-substituted enynes/enediynes would be applicable reaction partners for Lu's [3 + 2] cycloaddition since the resulting products can serve as useful building blocks for further post-functionalization through numerous metal- or organo-catalyzed reactions. Herein we present phosphine-catalyzed [3 + 2] cycloadditions of allenoates with CF_3-substituted enynes/enediynes, which give rise to alkyne- and diyne-tethered cyclopentenes bearing a CF_3-substituted quaternary carbon center (Scheme 1d).

Initial studies were carried out with allenoate **1a** and CF_3-substituted enyne **2a**. The expected [3 + 2] cycloaddition proceeded smoothly with excellent regioselectivity in the presence of 10 mol% PPh_3 in toluene at room temperature, giving α-adduct **3aa** in 76% yield (Table 1, entry 1). In order to increase the yield, various solvents were then examined, among which both THF and EtOAc gave improved yields while EtOH led to no reaction (entries 2–6). In view of its environmental benignity, EtOAc was selected as the optimal solvent for this reaction. Next, a number of commercially available mono- and bisphosphines were surveyed. Among those, $(4\text{-}MeOC_6H_4)_3P$ gave a better yield than $(4\text{-}FC_6H_4)_3P$ (entries 7 and 8). PPh_2Me afforded **3aa** in a yield comparable to that of PPh_3, while PBu_3 led to the rapid decomposition of the starting material (entries 9 and 10). Bisphosphines, such as DPPF, BINAP, and DPPP (1,3-bis(diphenylphosphino) propane), were either less effective than PPh_3 or ineffective (entries 11–13). When DABCO (1,4-diazabicyclo[2.2.2]octane) was used, the α-adduct **3aa** was obtained in 32% yield with >20 : 1 regioselectivity (entry 14).

Table 1 Optimization of the reaction conditions[a]



Entry	Catalyst	Solvent	Yield[b] **3aa** (%)	α/γ[c]
1	PPh_3	Toluene	76	>20 : 1
2	PPh_3	THF	87	>20 : 1
3	PPh_3	MeCN	31	>20 : 1
4	PPh_3	DCM	66	>20 : 1
5	PPh_3	EtOAc	89	>20 : 1
6	PPh_3	EtOH	Trace	—
7	$(4\text{-}FC_6H_4)_3P$	EtOAc	40	>20 : 1
8	$(4\text{-}MeOC_6H_4)_3P$	EtOAc	86	>20 : 1
9	PPh_2Me	EtOAc	84	>20 : 1
10	PBu_3	EtOAc	Trace	—
11	DPPF	EtOAc	82	>20 : 1
12	BINAP	EtOAc	Trace	—
13	DPPP	EtOAc	57	>20 : 1
14	DABCO	EtOAc	32	>20 : 1

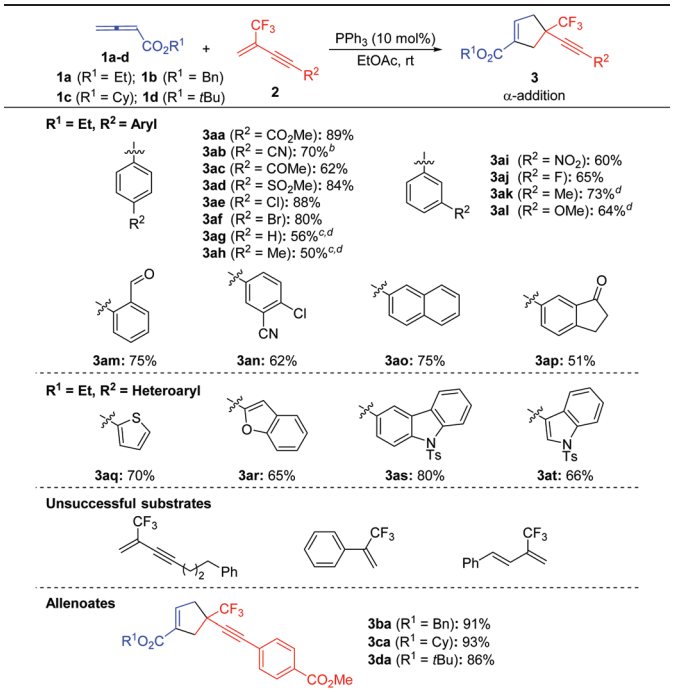
[a] Conditions: Reactions were performed with **1a** (0.4 mmol), **2a** (0.2 mmol) and the catalyst (10 mol%) in the solvent (2 mL) at rt for 24 h under N_2. [b] Isolated yield. [c] The α/γ ratio was determined by ^1H NMR analysis of the crude products.

Having established the optimal reaction conditions, the scope of this [3 + 2] cycloaddition was then probed (Table 2). First, various 2-trifluoromethyl-1,3-enynes were investigated. In general, the electronic nature of substituents on the aromatic rings had little effect on the regioselectivity but noticeable influences on the yields. The enynes bearing electron-withdrawing groups on the *para*- and *ortho*-positions of the phenyl ring exhibited higher reactivities toward this cycloaddition, giving the desired products (**3aa–3af**, **3am**) in moderate to good yields (62–88%). CF_3-Enynes with a phenyl or tolyl moiety turned out to be less reactive, affording products **3ag** and **3ah** in 56% and 50% yields, respectively, under modified conditions. Moreover, CF_3-enynes with the electron-withdrawing group (–NO_2 or –F) and the electron-donating group (–Me, –OMe) on the *meta*-position of the phenyl ring were all tolerant, and the corresponding products **3ai–3al** were obtained in 60–73% yields. In the case of disubstituted and fused-ring aromatic substrates, the desired products **3an–3ap** were produced in 51–75% yields. To our delight, substrates containing a heteroaromatic ring (thiophene, benzofuran, indole, and carbazole) on the alkyne moiety were also applicable, and the desired products (**3aq–3at**) were obtained in moderate to good yields. However, alkyl-substituted CF_3-enyne, as well as CF_3-styrene and CF_3-diene, failed to undergo this [3 + 2] cyclo-

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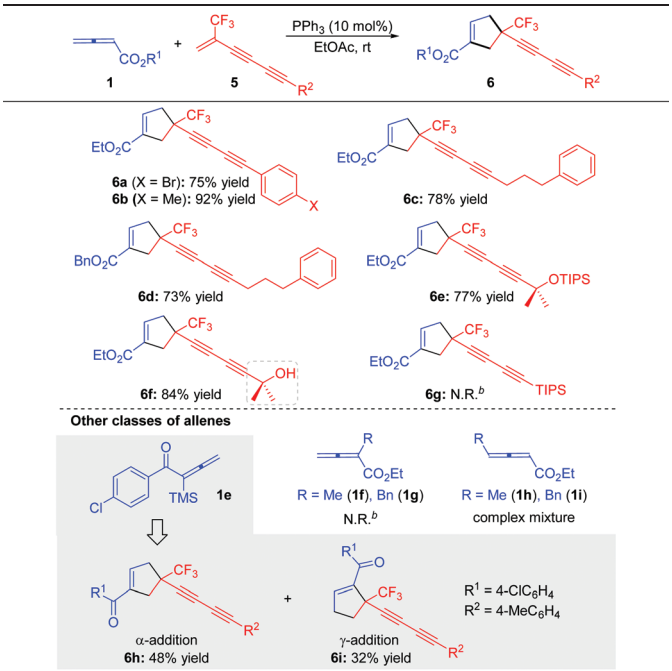
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Table 2 [3 + 2] cycloadditions of CF_3-enynes with allenoates[a]



[a] Conditions: Reactions were performed with **1** (0.4 mmol), **2** (0.2 mmol) and PPh_3 (10 mol%) in EtOAc (2 mL) at rt for 24 h under N_2; isolated yield; in all cases, the α/γ ratio is >20 : 1 unless otherwise noted. [b] α/γ = 16 : 1. [c] 0.6 mmol of **1a** was used in two portions. [d] 48 h.

Table 3 [3 + 2] cycloadditions of CF_3-enediynes with allenoates[a]

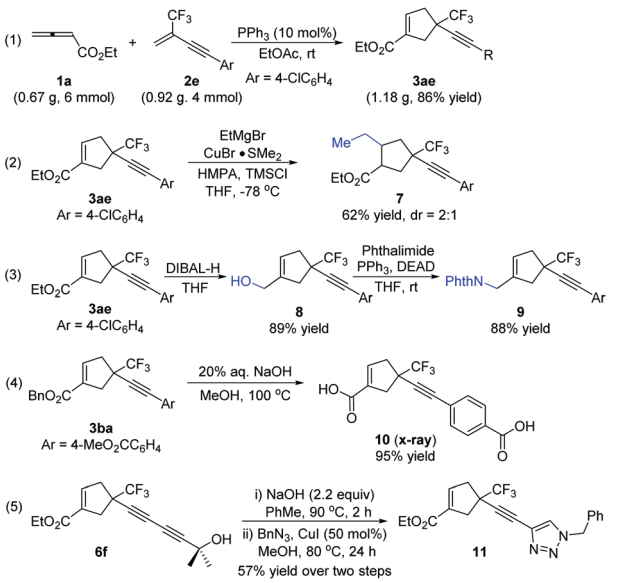


[a] Conditions: Reactions were performed with **1** (0.2 mmol), **5** (0.1 mmol) and PPh_3 (10 mol%) in EtOAc (1 mL) at rt for 48 h under N_2; isolated yield; in all cases, the α/γ ratio is >20 : 1. [b] N.R. = no reaction.

addition with **1a**. In addition, the substrate scope with respect to the allenoate was examined. Other allenoates, such as benzyl-, cyclohexyl-, and *tert*-butyl 2,3-butadienoates, proved to be suitable substrates, affording the corresponding products **3ba–3da** in 86–93% yields with an α/γ ratio of >20 : 1.

Encouraged by these results, we further extended our studies to the synthesis of diyne-tethered cyclopentenes *via* the cycloaddition of CF_3-enediynes (Table 3).[16] The CF_3-enediynes bearing an electron-withdrawing group (Br) and an electron-donating group (Me) on the *para*-position of the phenyl ring were applicable, and the desired products **6a** and **6b** were obtained in 75% and 92% yields, respectively. To our delight, alkyl CF_3-enediynes also proved to be competent substrates toward this cycloaddition, affording the corresponding adducts in good yields (**6c–6f**). Notably, the acetone-derived protecting groups were tolerated under the optimal conditions, which provide feasibility for further structural elaborations (**6e** and **6f**).[13b] However, TIPS-protected enediyne failed to give the target compound (**6g**). In addition, several different classes of allenes were also evaluated by using *p*-tolyl-substituted CF_3-enediyne **5b** as the reaction partner. The reaction occurred readily in the case of α-trimethylsilyl-substituted aryl allenone **1e**,[11] giving the desired α-cycloadduct **6h**, along with γ-cycloadduct **6i**, in 48% and 32% yields, respectively. The α-methyl/benzyl-substituted allenoates (**1f** and **1g**) proved to be completely unreactive, while the use of γ-methyl/phenyl-substituted allenoates (**1h** and **1i**) resulted in the formation of a complex mixture under standard conditions.

To showcase the practicability of this [3 + 2] cycloaddition, a gram-scale synthesis of CF_3-substituted cyclopentene **3ae** was performed. As illustrated in Scheme 2, the reaction was con-

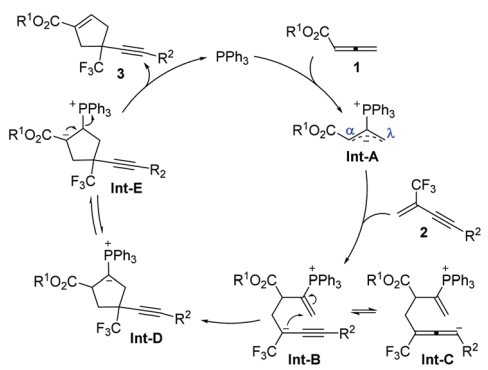


Scheme 2 Reaction on a gram scale and further structural elaborations.

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ducted with 2-trifluoromethyl-1,3-enyne **2e** (0.92 g, 4 mmol) with only 1.5 equivalents of allenoate **1a** (0.67 g, 6 mmol). The reaction proceeded smoothly, affording the corresponding product **3ae** with a yield (86%, 1.18 g) comparable to that of the small-scale reaction (Scheme 2, eqn (1)). The generated **3ae** could serve as a Michael acceptor undergoing Cu-catalysed Michael addition with ethylmagnesium bromide to provide cyclopentane **7** with 62% yield and a 2 : 1 dr value (Scheme 2, eqn (2)).[5a] The ester group of **3ae** could be selectively reduced to the alcohol in 89% yield with diisobutyl aluminium hydride (DIBAL-H), and the resulting allylic alcohol **8** could be further transferred into protected allylic amine **9** *via* the Mitsunobu reaction with phthalimide (Scheme 2, eqn (3)).[17] In addition, the hydrolyzation of cyclopentene **3ba** into dicarboxylic acid **10** was achieved in 95% yield (Scheme 2, eqn (4)). The structure of compound **10** was unambiguously confirmed by X-ray single-crystal analysis (CCDC 2004999†). Finally, the elaboration of product **6f** was performed to demonstrate the synthetic utility of the alkyne functionality. The acetone-derived protecting group on **6f** was removed by treating with NaOH, and the resulting terminal alkyne could further undergo azide–alkyne Huisgen cycloaddition, giving the 1,2,3-triazole **11** in 57% yield over two steps (Scheme 2, eqn (5)).

On the basis of the reported literature[18] and our experimental findings, a stepwise mechanism is proposed in Scheme 3. Initially, nucleophilic addition of phosphine onto allenoate **1** forms the reactive zwitterionic intermediate **Int-A**, which then undergoes α-addition to the double bond of 2-trifluoromethyl-1,3-enyne **2** to yield the propargylic anion intermediate **Int-B** or 1,2-allenic anion intermediate **Int-C**. Subsequent intramolecular conjugate addition of **Int-B** occurs to form the cyclic ylide intermediate **Int-D**. Finally, a [1,2] proton transfer followed by β-elimination of the catalyst gives rise to the target cyclopentene **3** with the concurrent release of PPh_3. The introduction of acetylene was key for a number of reasons: (a) the electron-withdrawing polarization of ethynyl groups may further activate CF_3-enynes for nucleophilic addition by the zwitterionic intermediate[19] and (b) ethynyl groups may help to stabilize the α-trifluoromethyl carbanion species electronically, thus facilitating the ring closure process.[20]



Scheme 3 Proposed reaction mechanism.

Conclusions

In conclusion, we have described the first phosphine-catalyzed [3 + 2] cycloaddition of 2-trifluoromethyl enynes/enediynes with allenoates, affording cyclopentenes containing a CF_3-substituted quaternary carbon center in moderate to high yields and with excellent regioselectivity. Furthermore, several elaborations based on the generated products were conducted to demonstrate their synthetic potential. This methodology successfully expanded the substrate scope of Lu's [3 + 2] cycloaddition with respect to electron-deficient olefins. The readily available catalyst, environment friendly solvent, and excellent regioselectivity make it a practical and expedient approach for accessing biologically important trifluoromethylated cyclopentenes.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We gratefully acknowledge the funding support from the National Natural Science Foundation of China (21801050, 21905089) and the Scientific Research Project of Guangzhou Municipal Colleges and Universities (No. 201831816). We thank Dr Wen-Bin Chen for X-ray diffraction analysis.

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